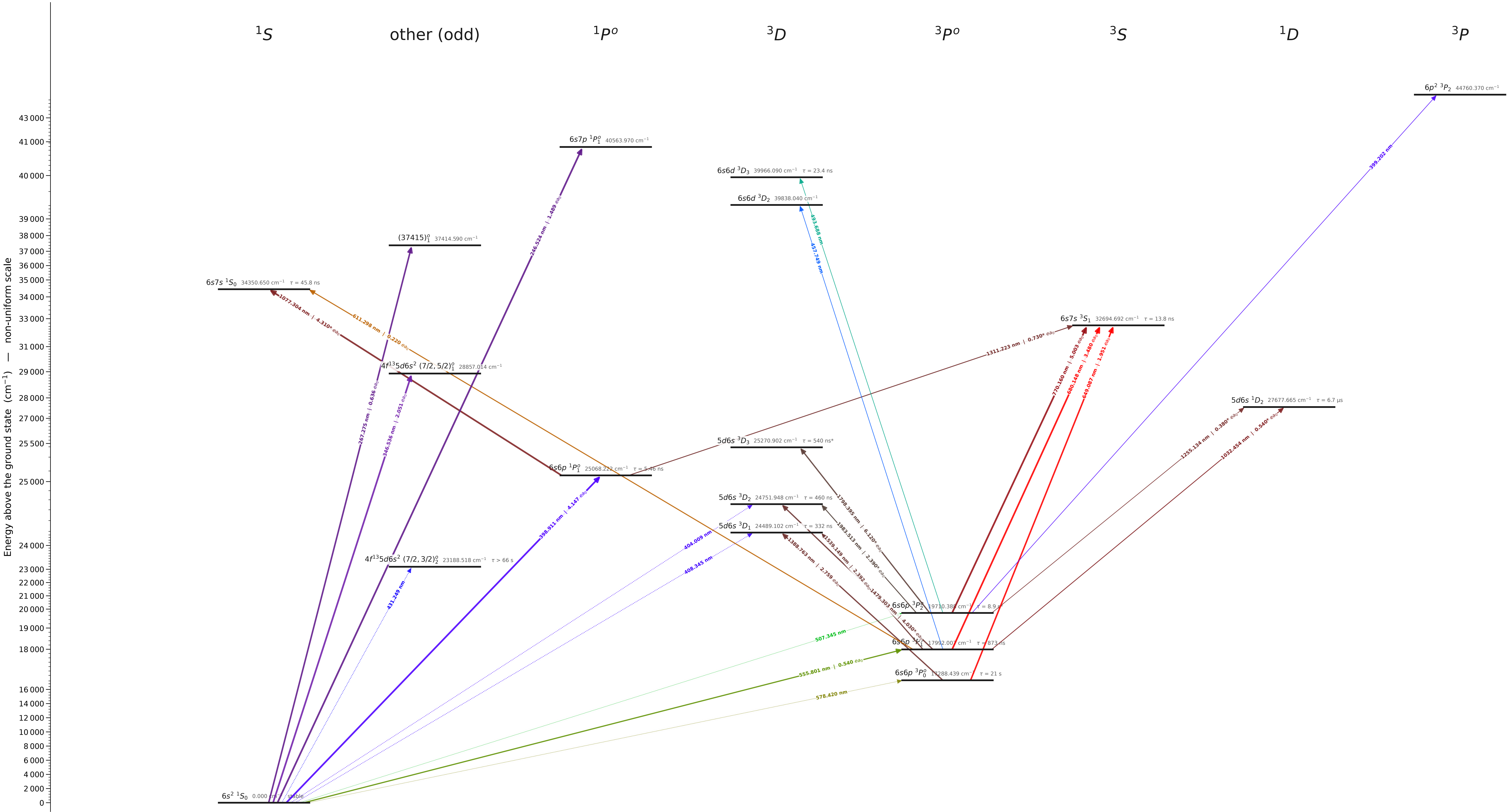
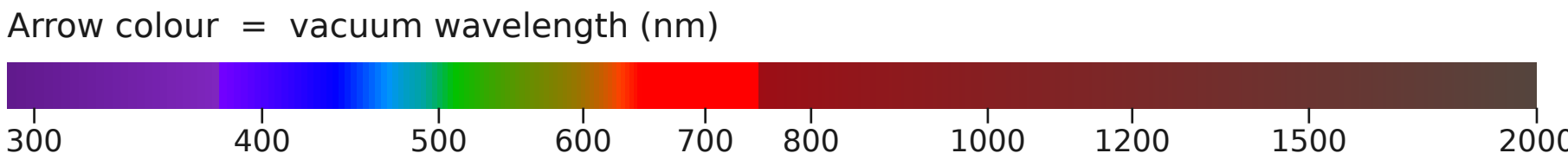


# Ytterbium-171

<sup>171</sup>Yb (Yb I), nuclear spin *I* = 1/2. Energy levels and transitions below 2 μm: 18 levels, 26 lines.

Yb <sup>+</sup> ionisation limit 50443.20 cm<sup>-1</sup> = 6.25416 eV (198.243 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element  $| \langle U || e r || J' \rangle |$  in *ea*<sub>0</sub>  
Matrix elements are measured or NIST values; \* = high-accuracy theory, ≈ = model calculation.  
Arrow width grows with the Einstein coefficient *A*. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, *n* = 70.  
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



## Key transitions

| Transition                                                                                                            | λ vac (nm) | λ air (nm) | ν (THz)      | ⟨U  J'⟩  (ea <sub>0</sub> ) | A (s <sup>-1</sup> ) | Γ/2π up. (MHz) | branch (%) | Use                                                          |
|-----------------------------------------------------------------------------------------------------------------------|------------|------------|--------------|-----------------------------|----------------------|----------------|------------|--------------------------------------------------------------|
| 6s <sup>2</sup> <sup>1</sup> S <sub>0</sub> – 6s7p <sup>1</sup> P <sub>1</sub> <sup>o</sup>                           | 246.5242   | 246.4497   | 1216.0772    | 1.489                       | 1.000e+08            | -              | -          | 246.5 nm                                                     |
| 6s <sup>2</sup> <sup>1</sup> S <sub>0</sub> – (37415) <sub>1</sub> <sup>o</sup>                                       | 267.2754   | 267.1960   | 1121.6612    | 0.636                       | 1.430e+07            | -              | -          | 267.3 nm                                                     |
| 6s <sup>2</sup> <sup>1</sup> S <sub>0</sub> – 4f <sup>13</sup> 5d6s <sup>2</sup> (7/2, 5/2) <sub>1</sub> <sup>o</sup> | 346.5362   | 346.4370   | 865.1115     | 2.051                       | 6.830e+07            | -              | -          | 346.5 nm                                                     |
| 6s <sup>2</sup> <sup>1</sup> S <sub>0</sub> – 6s6p <sup>1</sup> P <sub>1</sub> <sup>o</sup>                           | 398.9108   | 398.7981   | 751.527471   | 4.147                       | 1.830e+08            | 29.13          | 100        | 399 nm: Zeeman slower, blue MOT, imaging                     |
| 6s <sup>2</sup> <sup>1</sup> S <sub>0</sub> – 5d6s <sup>3</sup> D <sub>2</sub>                                        | 404.0086   | 403.8945   | 742.0447     | -                           | -                    | 0.346          | -          | 404 nm forbidden transition                                  |
| 6s <sup>2</sup> <sup>1</sup> S <sub>0</sub> – 5d6s <sup>3</sup> D <sub>1</sub>                                        | 408.3449   | 408.2296   | 734.1648     | -                           | -                    | 0.4794         | -          | 408 nm highly forbidden                                      |
| 6s <sup>2</sup> <sup>1</sup> S <sub>0</sub> – 4f <sup>13</sup> 5d6s <sup>2</sup> (7/2, 3/2) <sub>2</sub> <sup>o</sup> | 431.2489   | 431.1276   | 695.172740   | -                           | -                    | 2.411e-09      | -          | 431 nm inner-shell clock transition                          |
| 6s <sup>2</sup> <sup>1</sup> S <sub>0</sub> – 6s6p <sup>3</sup> P <sub>2</sub> <sup>o</sup>                           | 507.3452   | 507.2038   | 590.90433(4) | -                           | 1.120e-01            | 1.788e-08      | 99.7       | 507 nm M2 transition: shelving into 3P2                      |
| 6s <sup>2</sup> <sup>1</sup> S <sub>0</sub> – 6s6p <sup>3</sup> P <sub>1</sub> <sup>o</sup>                           | 555.8007   | 555.6464   | 539.388427   | 0.540                       | 1.146e+06            | 0.1824         | 100        | 556 nm: green MOT, narrow-line cooling, imaging in tweeze... |
| 6s <sup>2</sup> <sup>1</sup> S <sub>0</sub> – 6s6p <sup>3</sup> P <sub>0</sub> <sup>o</sup>                           | 578.4196   | 578.2592   | 518.295837   | -                           | 4.800e-02            | 7.579e-09      | 100        | 578 nm optical clock transition                              |
| 6s6p <sup>3</sup> P <sub>2</sub> <sup>o</sup> – 6s7s <sup>1</sup> S <sub>0</sub>                                      | 611.2976   | 611.1285   | 490.4198     | 0.220                       | 4.293e+05            | 3.475          | 1.97       | 611 nm                                                       |
| 6s6p <sup>3</sup> P <sub>0</sub> <sup>o</sup> – 6s7s <sup>3</sup> S <sub>1</sub>                                      | 649.0866   | 648.9073   | 461.868177   | 1.951                       | 9.400e+06            | 11.53          | 13         | 649 nm clock-state repumper / 3P0 detection                  |
| 6s6p <sup>3</sup> P <sub>1</sub> <sup>o</sup> – 6s7s <sup>3</sup> S <sub>1</sub>                                      | 680.1479   | 679.9602   | 440.7754     | 3.480                       | 2.600e+07            | 11.53          | 36         | 680 nm decay channel of 3S1 during repumping                 |
| 6s6p <sup>3</sup> P <sub>2</sub> <sup>o</sup> – 6s7s <sup>3</sup> S <sub>1</sub>                                      | 770.1603   | 769.9483   | 389.259834   | 5.003                       | 3.700e+07            | 11.53          | 50.8       | 770 nm 3P2 repumper                                          |
| 6s6p <sup>3</sup> P <sub>1</sub> <sup>o</sup> – 5d6s <sup>1</sup> D <sub>2</sub>                                      | 1032.4544  | 1032.1715  | 290.3687     | 0.540*                      | 1.074e+05            | 0.02375        | 71.9       | 1032 nm                                                      |
| 6s6p <sup>1</sup> P <sub>1</sub> <sup>o</sup> – 6s7s <sup>1</sup> S <sub>0</sub>                                      | 1077.3043  | 1077.0093  | 278.2802     | 4.310*                      | 3.010e+07            | 3.475          | 100        | 1077 nm main decay channel of 6s7s 1S0                       |
| 6s6p <sup>3</sup> P <sub>2</sub> <sup>o</sup> – 5d6s <sup>3</sup> D <sub>2</sub>                                      | 1255.1340  | 1254.7907  | 238.8530     | 0.380*                      | 2.959e+04            | 0.02375        | 19.8       | 1255 nm                                                      |
| 6s6p <sup>1</sup> P <sub>1</sub> <sup>o</sup> – 6s7s <sup>3</sup> S <sub>1</sub>                                      | 1311.2226  | 1310.8641  | 228.6358     | 0.730*                      | 1.596e+05            | 11.53          | 0.21       | 1311 nm weak decay channel of 3S1                            |
| 6s6p <sup>3</sup> P <sub>0</sub> <sup>o</sup> – 5d6s <sup>3</sup> D <sub>1</sub>                                      | 1388.7630  | 1388.3833  | 215.870145   | 2.759                       | 1.920e+06            | 0.4794         | 63.8       | 1389 nm clock-state repumper                                 |
| 6s6p <sup>3</sup> P <sub>1</sub> <sup>o</sup> – 5d6s <sup>3</sup> D <sub>2</sub>                                      | 1479.3029  | 1478.8986  | 202.6579     | 4.030*                      | 2.033e+06            | 0.346          | 93.5       | 1480 nm                                                      |
| 6s6p <sup>3</sup> P <sub>1</sub> <sup>o</sup> – 5d6s <sup>3</sup> D <sub>1</sub>                                      | 1539.1494  | 1538.7289  | 194.7780     | 2.392                       | 1.060e+06            | 0.4794         | 35.2       | 1539 nm decay channel of 3D1                                 |
| 6s6p <sup>3</sup> P <sub>2</sub> <sup>o</sup> – 5d6s <sup>3</sup> D <sub>3</sub>                                      | 1798.3949  | 1797.9039  | 166.7000     | 6.120*                      | 1.864e+06            | 0.2947         | 100        | 1798 nm cycling transition from 3P2                          |
| 6s6p <sup>3</sup> P <sub>2</sub> <sup>o</sup> – 5d6s <sup>3</sup> D <sub>2</sub>                                      | 1983.5130  | 1982.9717  | 151.1422     | 2.390*                      | 2.966e+05            | 0.346          | 13.6       | 1983 nm                                                      |

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.  
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

## <sup>171</sup>Yb hyperfine structure (*I* = 1/2)

| Level                                                                   | A (MHz)      | B (MHz) | Shift of each F level from the centre of gravity (MHz) |
|-------------------------------------------------------------------------|--------------|---------|--------------------------------------------------------|
| 6s6p <sup>3</sup> P <sub>1</sub> <sup>o</sup>                           | 3957.85(5)   | -       | F=1/2: -3957.85 F=3/2: +1978.92                        |
| 6s6p <sup>3</sup> P <sub>2</sub> <sup>o</sup>                           | 2681.4(26)   | -       | F=3/2: -4022.10 F=5/2: +2681.40                        |
| 4f <sup>13</sup> 5d6s <sup>2</sup> (7/2, 3/2) <sub>2</sub> <sup>o</sup> | 1123.354(13) | -       | F=3/2: -1685.03 F=5/2: +1123.35                        |
| 5d6s <sup>3</sup> D <sub>1</sub>                                        | -2038.30(11) | -       | F=1/2: +2038.30 F=3/2: -1019.15                        |
| 5d6s <sup>3</sup> D <sub>2</sub>                                        | 1314(3)      | -       | F=3/2: -1970.70 F=5/2: +1313.80                        |
| 6s6p <sup>1</sup> P <sub>1</sub> <sup>o</sup>                           | -214.06(17)  | -       | F=1/2: +214.06 F=3/2: -107.03                          |
| 6s7s <sup>3</sup> S <sub>1</sub>                                        | 6838.41(26)  | -       | F=1/2: -6838.41 F=3/2: +3419.20                        |

Measured A, B constants; sources in the data file.

## Isotope shifts

| Transition                                                                                                            | λ vac (nm) | Isotopes | Shift (MHz)   |
|-----------------------------------------------------------------------------------------------------------------------|------------|----------|---------------|
| 6s <sup>2</sup> <sup>1</sup> S <sub>0</sub> – 6s6p <sup>1</sup> P <sub>1</sub> <sup>o</sup>                           | 398.911    | 168-174  | 1887.64(23)   |
| 6s <sup>2</sup> <sup>1</sup> S <sub>0</sub> – 6s6p <sup>1</sup> P <sub>1</sub> <sup>o</sup>                           | 398.911    | 170-174  | 1187.8(27)    |
| 6s <sup>2</sup> <sup>1</sup> S <sub>0</sub> – 6s6p <sup>1</sup> P <sub>1</sub> <sup>o</sup>                           | 398.911    | 171-174  | 940.8(9)      |
| 6s <sup>2</sup> <sup>1</sup> S <sub>0</sub> – 6s6p <sup>1</sup> P <sub>1</sub> <sup>o</sup>                           | 398.911    | 172-174  | 533.0(3)      |
| 6s <sup>2</sup> <sup>1</sup> S <sub>0</sub> – 6s6p <sup>1</sup> P <sub>1</sub> <sup>o</sup>                           | 398.911    | 173-174  | 291.51(5)     |
| 6s <sup>2</sup> <sup>1</sup> S <sub>0</sub> – 6s6p <sup>1</sup> P <sub>1</sub> <sup>o</sup>                           | 398.911    | 176-174  | -509.25(7)    |
| 6s <sup>2</sup> <sup>1</sup> S <sub>0</sub> – 5d6s <sup>3</sup> D <sub>2</sub>                                        | 404.009    | 170-174  | 2531(5)       |
| 6s <sup>2</sup> <sup>1</sup> S <sub>0</sub> – 5d6s <sup>3</sup> D <sub>2</sub>                                        | 404.009    | 171-174  | 2019(6)       |
| 6s <sup>2</sup> <sup>1</sup> S <sub>0</sub> – 5d6s <sup>3</sup> D <sub>2</sub>                                        | 404.009    | 172-174  | 1100(4)       |
| 6s <sup>2</sup> <sup>1</sup> S <sub>0</sub> – 5d6s <sup>3</sup> D <sub>2</sub>                                        | 404.009    | 173-174  | 609(6)        |
| 6s <sup>2</sup> <sup>1</sup> S <sub>0</sub> – 5d6s <sup>3</sup> D <sub>2</sub>                                        | 404.009    | 176-174  | -1059(11)     |
| 6s <sup>2</sup> <sup>1</sup> S <sub>0</sub> – 5d6s <sup>3</sup> D <sub>1</sub>                                        | 408.345    | 170-174  | 2542(12)      |
| 6s <sup>2</sup> <sup>1</sup> S <sub>0</sub> – 5d6s <sup>3</sup> D <sub>1</sub>                                        | 408.345    | 171-174  | 2038.9(12)    |
| 6s <sup>2</sup> <sup>1</sup> S <sub>0</sub> – 5d6s <sup>3</sup> D <sub>1</sub>                                        | 408.345    | 172-174  | 1112(8)       |
| 6s <sup>2</sup> <sup>1</sup> S <sub>0</sub> – 5d6s <sup>3</sup> D <sub>1</sub>                                        | 408.345    | 173-174  | 619.4(8)      |
| 6s <sup>2</sup> <sup>1</sup> S <sub>0</sub> – 5d6s <sup>3</sup> D <sub>1</sub>                                        | 408.345    | 176-174  | -1063(6)      |
| 6s <sup>2</sup> <sup>1</sup> S <sub>0</sub> – 4f <sup>13</sup> 5d6s <sup>2</sup> (7/2, 3/2) <sub>2</sub> <sup>o</sup> | 431.249    | 171-176  | -3406.79(3)   |
| 6s <sup>2</sup> <sup>1</sup> S <sub>0</sub> – 4f <sup>13</sup> 5d6s <sup>2</sup> (7/2, 3/2) <sub>2</sub> <sup>o</sup> | 431.249    | 173-176  | -1797.802(26) |
| 6s <sup>2</sup> <sup>1</sup> S <sub>0</sub> – 4f <sup>13</sup> 5d6s <sup>2</sup> (7/2, 3/2) <sub>2</sub> <sup>o</sup> | 431.249    | 174-176  | -1115.787(25) |
| 6s <sup>2</sup> <sup>1</sup> S <sub>0</sub> – 4f <sup>13</sup> 5d6s <sup>2</sup> (7/2, 3/2) <sub>2</sub> <sup>o</sup> | 431.249    | 172-176  | -2296.403(25) |
| 6s <sup>2</sup> <sup>1</sup> S <sub>0</sub> – 4f <sup>13</sup> 5d6s <sup>2</sup> (7/2, 3/2) <sub>2</sub> <sup>o</sup> | 431.249    | 170-176  | -3926.431(26) |
| 6s <sup>2</sup> <sup>1</sup> S <sub>0</sub> – 4f <sup>13</sup> 5d6s <sup>2</sup> (7/2, 3/2) <sub>2</sub> <sup>o</sup> | 431.249    | 168-176  | -5680.383(25) |

Shift = ν(first isotope) – ν(second isotope).